

Seneca

SEG800

Digital Signal and Image Processing

Introduction

Seneca College

Overview

- Definition of Signals and Systems
- Classification of Signals
- Discrete-time Signals
 - Representations and Transforms
- Signal Properties

Why DSP?

- Signal Processing has many applications.
- Due to advances in computing and integrated circuits (IC), digital signal processing is more common now.
- Other alternatives: analog signal processing, optical signal processing
- Question: in which applications is DSP not an option?

Signals

- Signal: a function of one or more independent variables.
- What is (are) the independent variable(s) in the following functions?

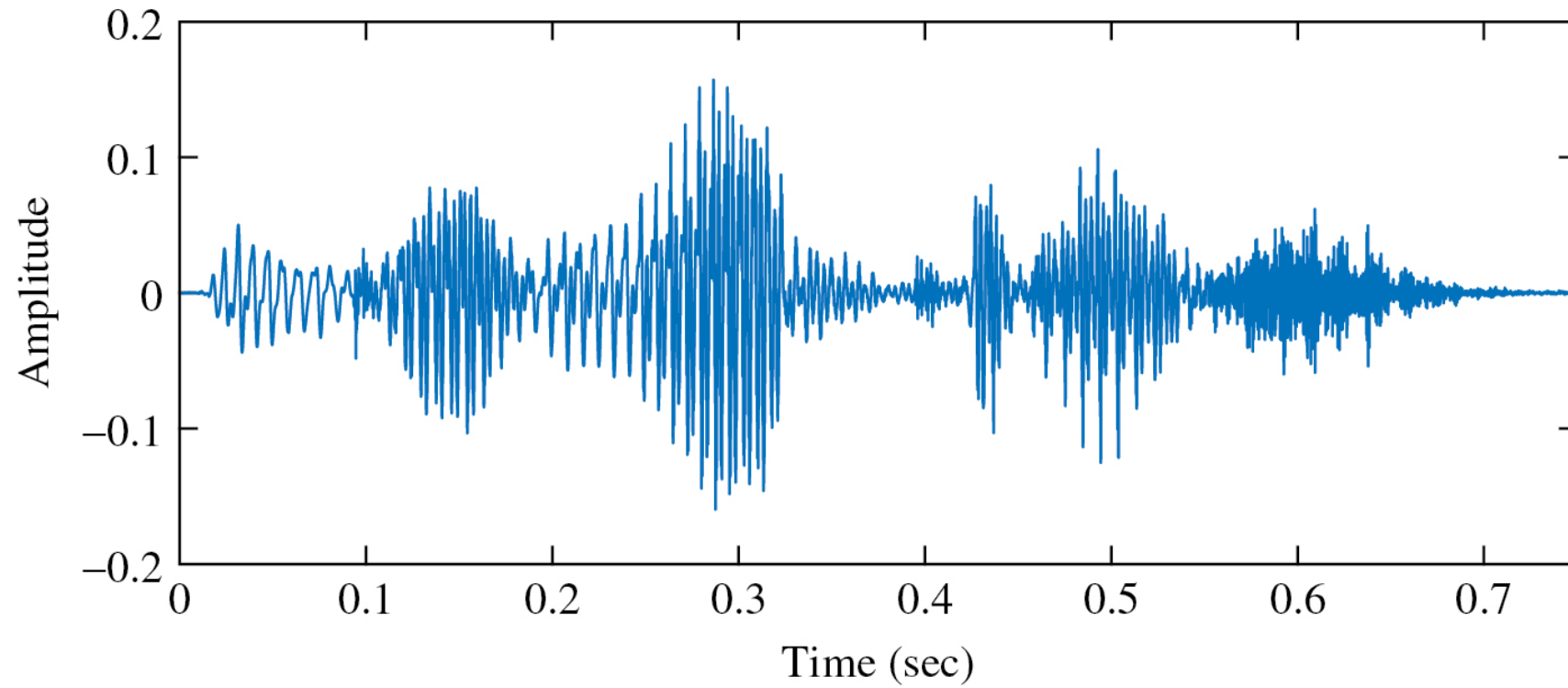
$$s_1(t) = 5t$$

$$s_2(t) = 20t^2$$

$$s(x, y) = 3x + 2xy + 10y^2$$

$$\sum_{i=1}^N A_i(t) \sin[2\pi F_i(t)t + \theta_i(t)]$$

Figure 1.1.1 Example of a speech signal.



Systems

- System: a software or physical device (hardware) that performs an operation on a signal.
 - Example: a noise reduction system (filter)
 - Linear systems
 - Nonlinear Systems
 - Such operations are usually referred to as **Signal Processing**.
- Algorithm: The method or set of rules for implementing the system by a program that performs the corresponding mathematical operations.

Figure 1.1.2 Analog signal processing.

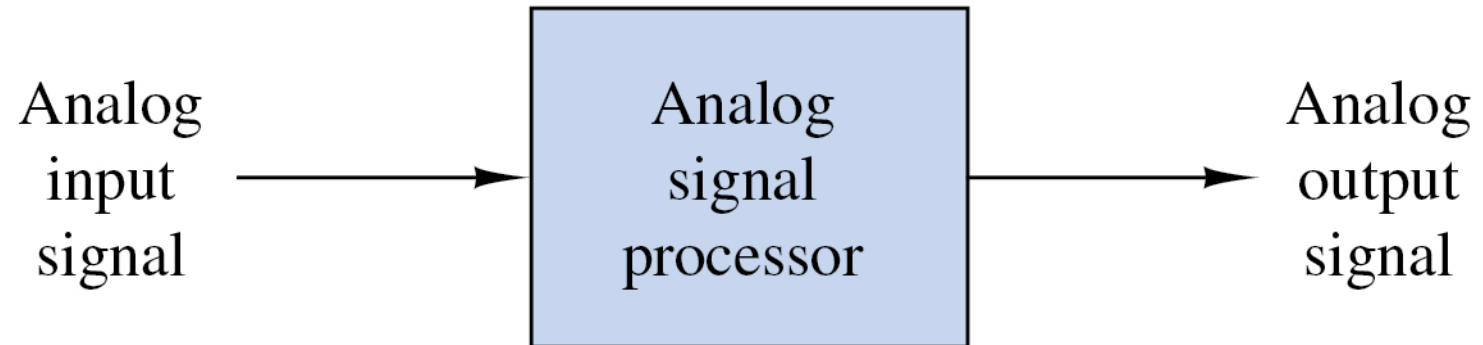
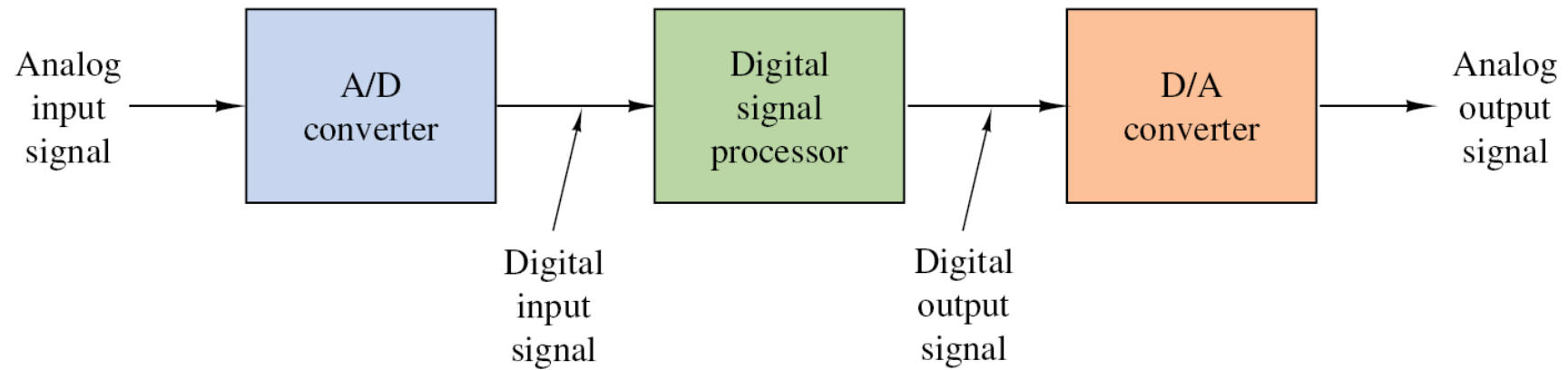


Figure 1.1.3 Block diagram of a digital signal processing system.

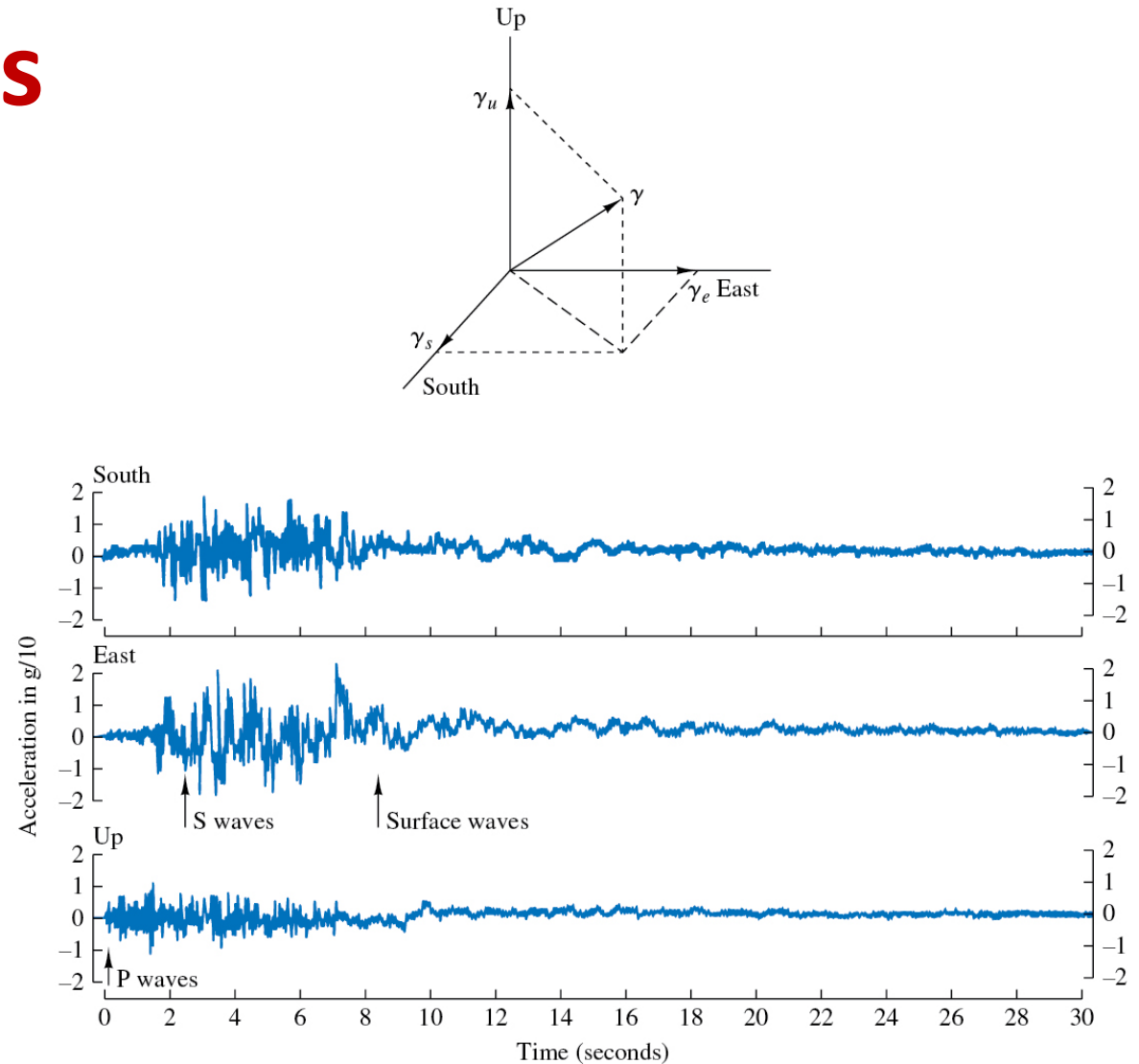


Advantages of DSP over Analog

- Lower Cost
- Speed
- Reliability
- Accuracy and higher precision
- Programmability and flexibility

Multichannel signals

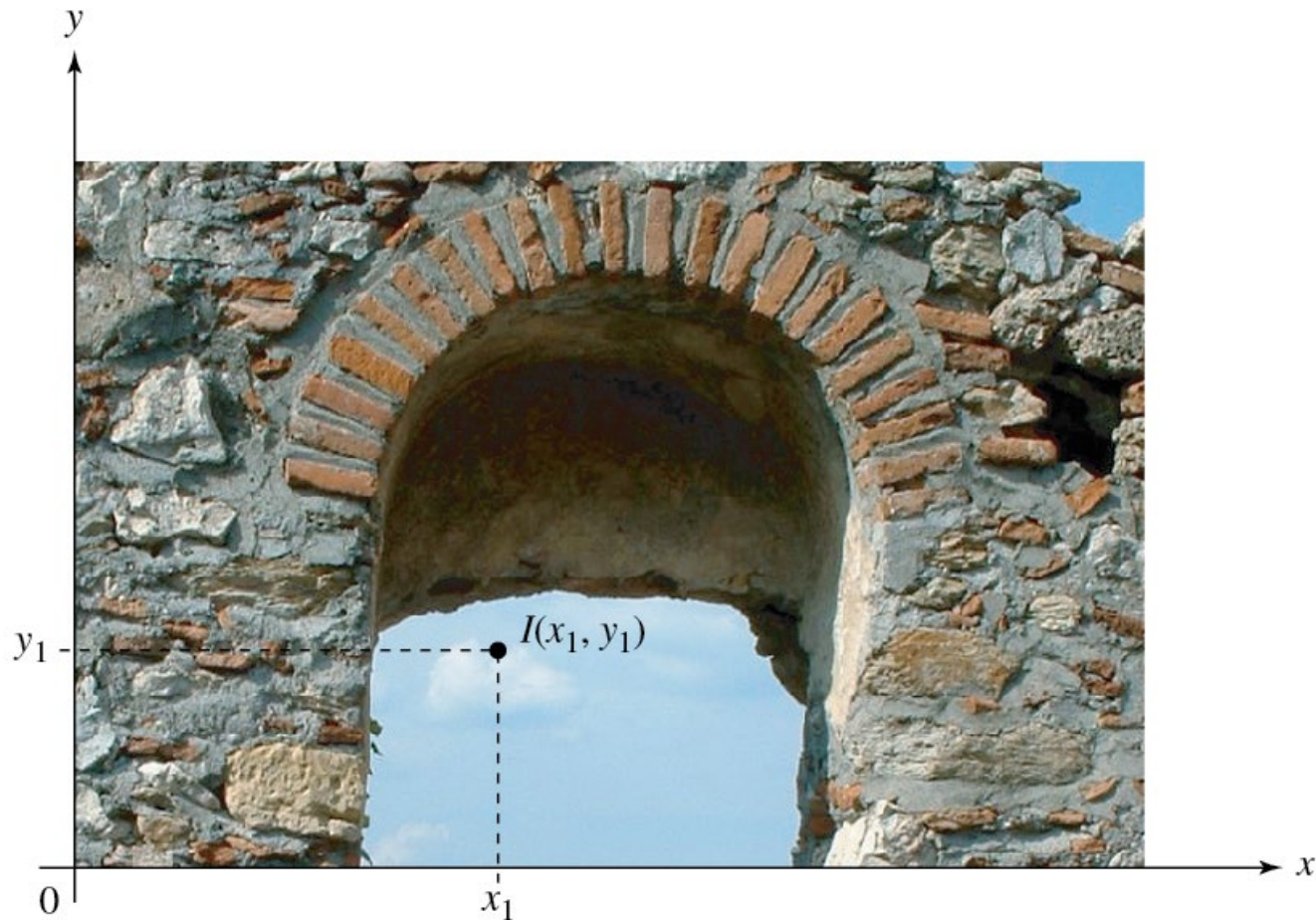
Figure 1.2.1 Three components of ground acceleration measured a few kilometers from the epicenter of an earthquake.



(From *Earthquakes*, by B. A. Bold, ©1988 by W. H. Freeman and Company.
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Multidimensional signals

Figure 1.2.2 Example of a two-dimensional signal.



Continuous-Time vs. Discrete-Time

- Continuous-time signals (**analog**) signals: defined for every value of time, with values in the continuous interval (a,b), where a can be $-\infty$ and b can be ∞ .

- Examples: $x_1(t) = \cos \pi t, x_2(t) = e^{-|t|}, -\infty < t < \infty$

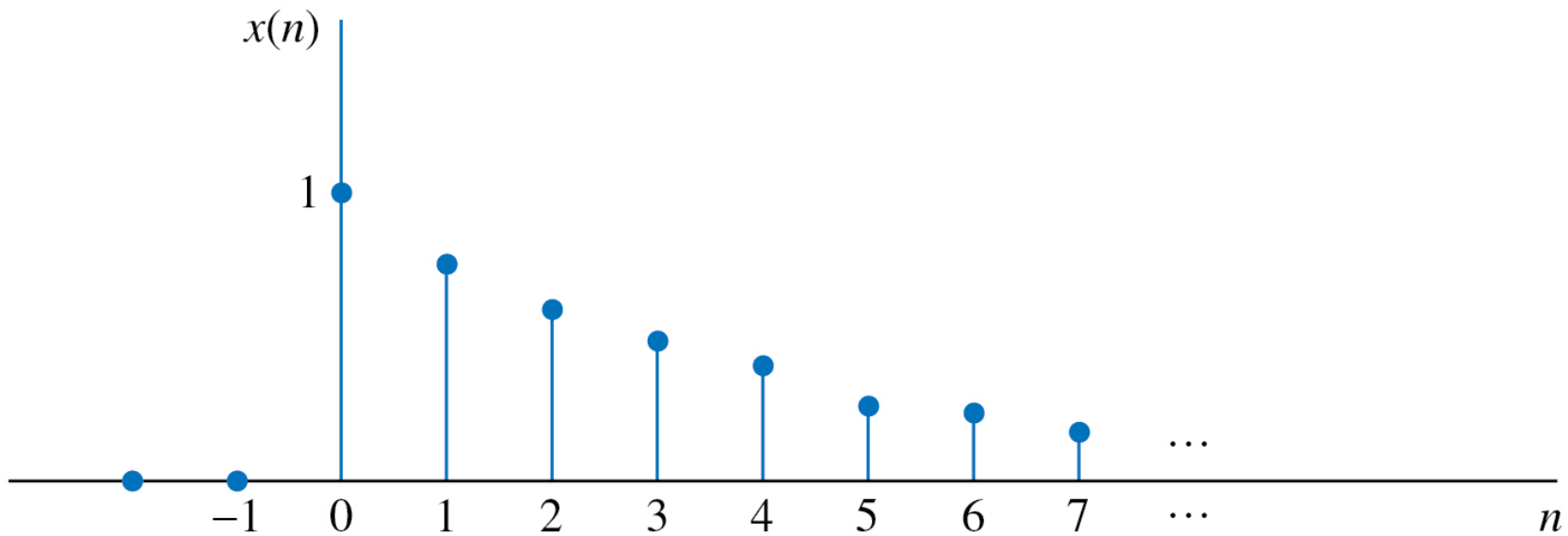
- Discrete-time signals: defined only at certain specific values of time. (Undefined otherwise)

- Examples:

$$x(t_n) = e^{-|t_n|}, n = 0, \pm 1, \pm 2, \dots \quad x(n) = \begin{cases} 0.8^n, & \text{if } n \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

- **Sampling:** Selecting values of an analog signal at discrete-times.

Figure 1.2.3 Graphical representation of the discrete time signal $x(n] = 0.8^n$ for $n > 0$ and $x(n] = 0$ for $n < 0$.



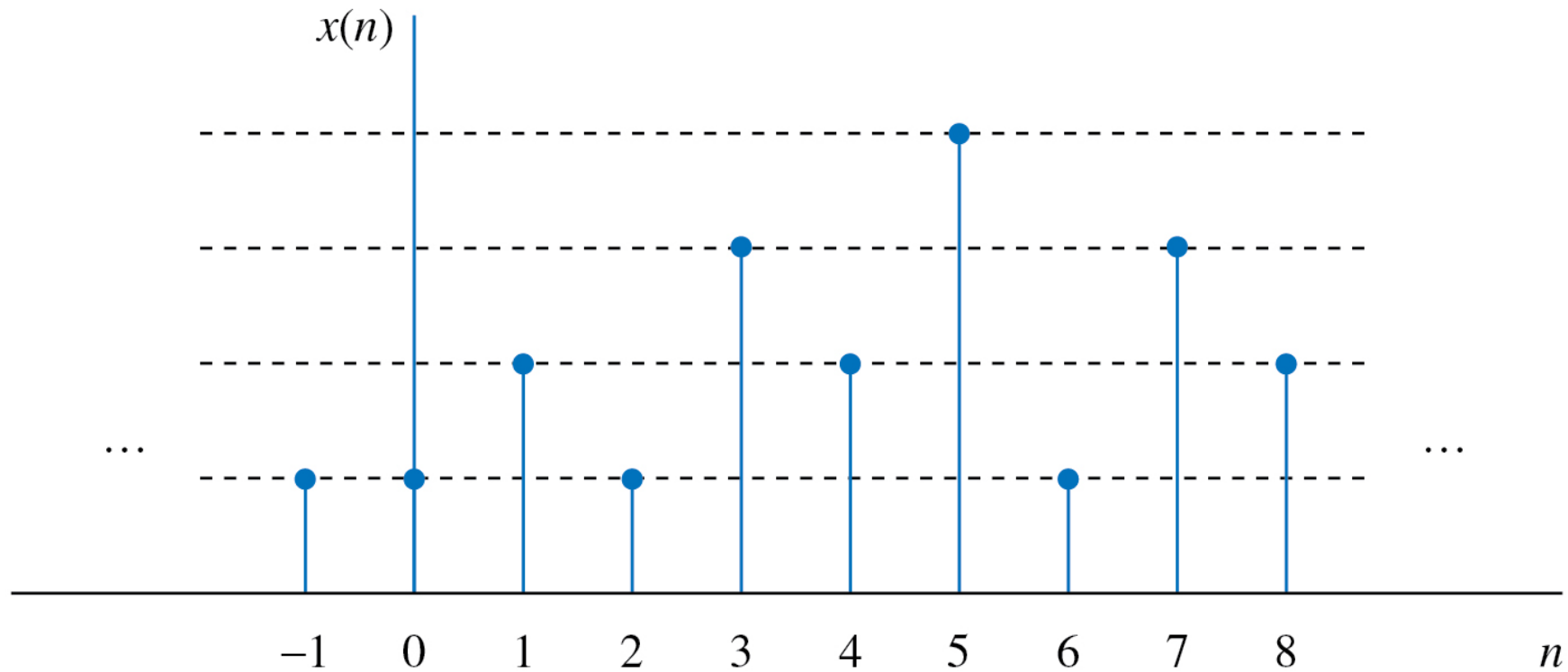
Continuous-Valued vs. Discrete-Valued

- Continuous-valued: any real number.
- Discrete-valued: discrete values (as saved and presented in a digital computer).
- Digital Signal: A discrete-time and discrete-valued signal.
- **Quantization:** The process of converting a continuous-valued signal into a discrete-valued signal.
 - Rounding or truncation are simple quantization methods.

Deterministic vs. Random Signals

- Deterministic: Any signal that can be uniquely described by an explicit mathematical expression, a table of data, or a well-defined rule.
 - All past, present, and future values of the signal are known precisely, without any uncertainty.
- Random: Signals that cannot be described by explicit mathematical formulas (or if such a description is too complicated or inaccurate).
 - Unpredictable or uncertain.

Figure 1.2.5 Digital signal with four different amplitude values.



Discrete-Time Signals

Representation of Discrete-Time Signals

- Representation:

- Functional

$$x(n) = \begin{cases} 1, & \text{for } n = 1, 3 \\ 4, & \text{for } n = 2 \\ 0, & \text{elsewhere} \end{cases}$$

- Tabular

n	$\dots$	-2	-1	0	1	2	3	4	5	$\dots$
$x(n)$	$\dots$	0	0	0	1	4	1	0	0	$\dots$

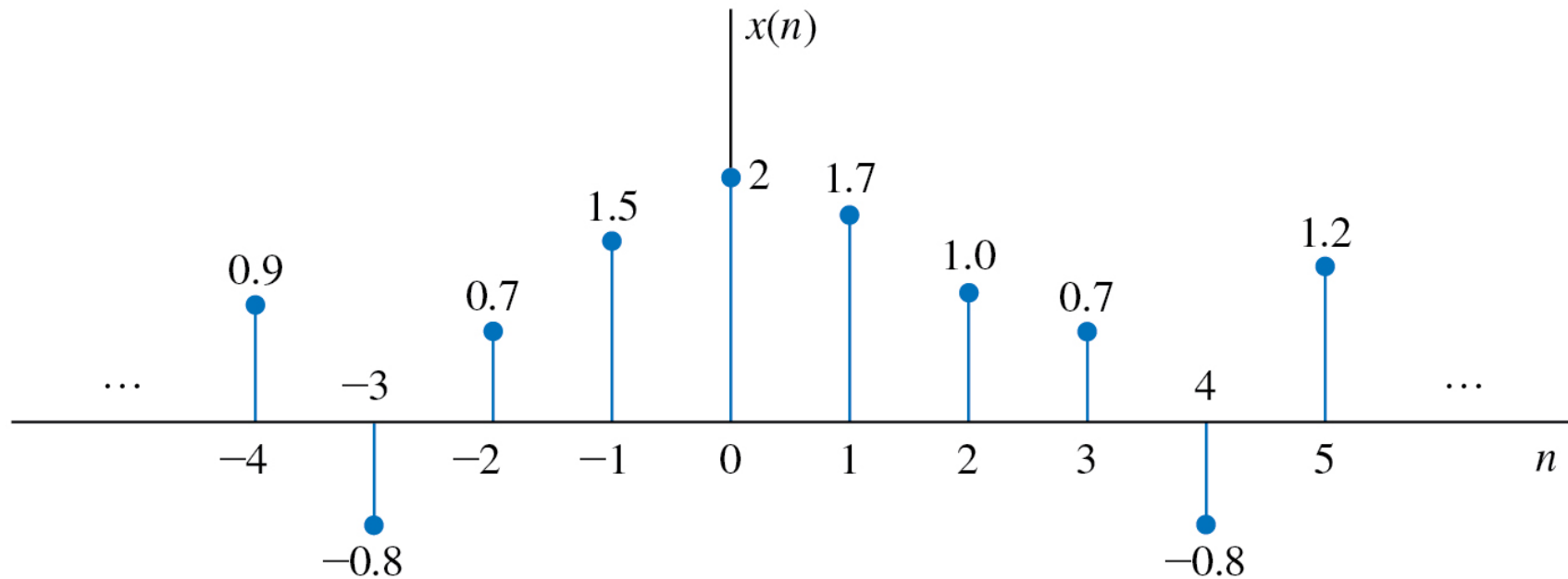
- Sequence

$$x(n) = \{\dots 0, \underset{\uparrow}{0}, 1, 4, 1, 0, 0, \dots\}$$

$$x(n) = \{3, -1, -\underset{\uparrow}{2}, 5, 0, 4, -1\}$$

- Graphical

Figure 2.1.1 Graphical representation of a discrete-time signal.



Elementary Discrete-Time Signals

- Unit sample sequence (or unit impulse) $\delta(n) \equiv \begin{cases} 1, & \text{for } n = 0 \\ 0, & \text{for } n \neq 0 \end{cases}$
- Unit step signal $u(n) \equiv \begin{cases} 1, & \text{for } n \geq 0 \\ 0, & \text{for } n < 0 \end{cases}$
- Unit ramp signal $u_r(n) \equiv \begin{cases} n, & \text{for } n \geq 0 \\ 0, & \text{for } n < 0 \end{cases}$
- Exponential signal $x(n) = a^n \quad \text{for all } n$

Figure 2.1.2 Graphical representation of the unit sample signal.

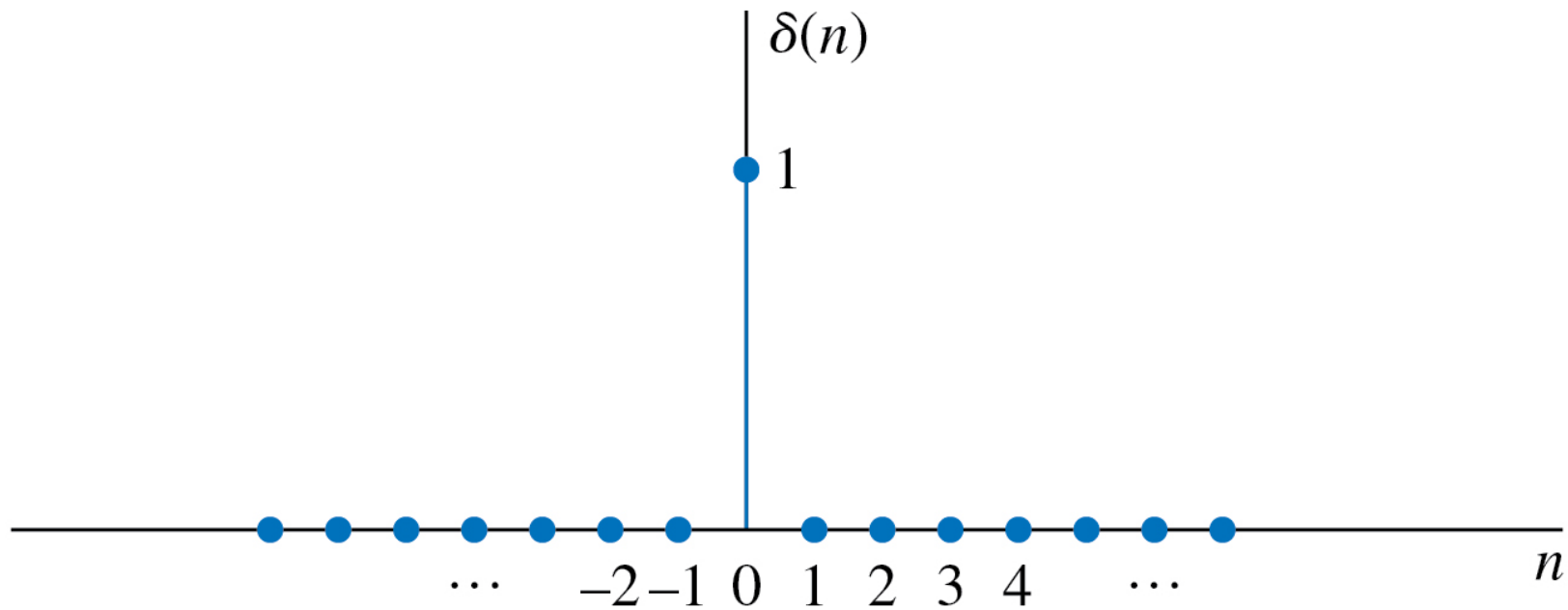


Figure 2.1.3 Graphical representation of the unit step signal.

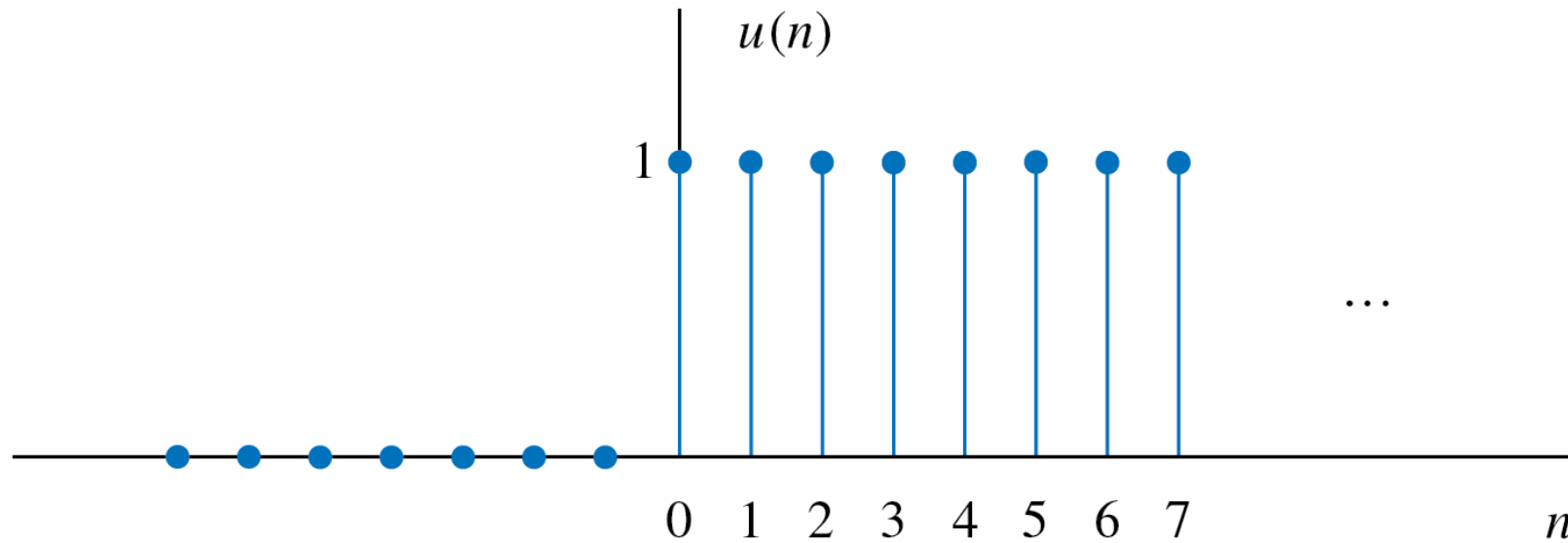


Figure 2.1.4 Graphical representation of the unit ramp signal.

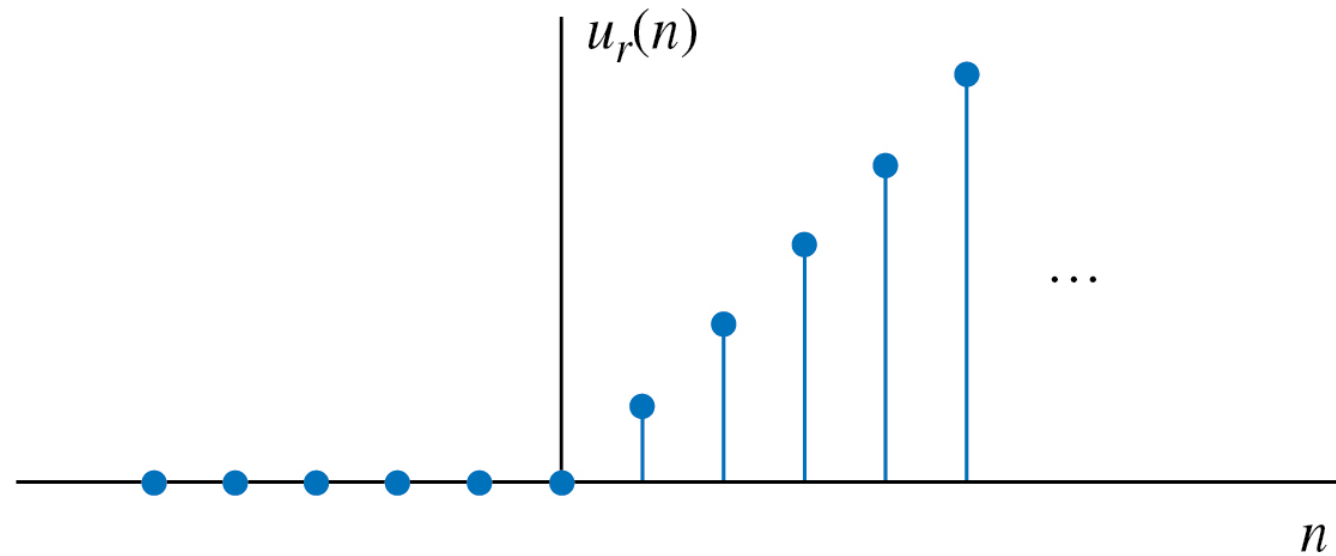


Figure 2.1.5 Graphical representation of exponential signal.

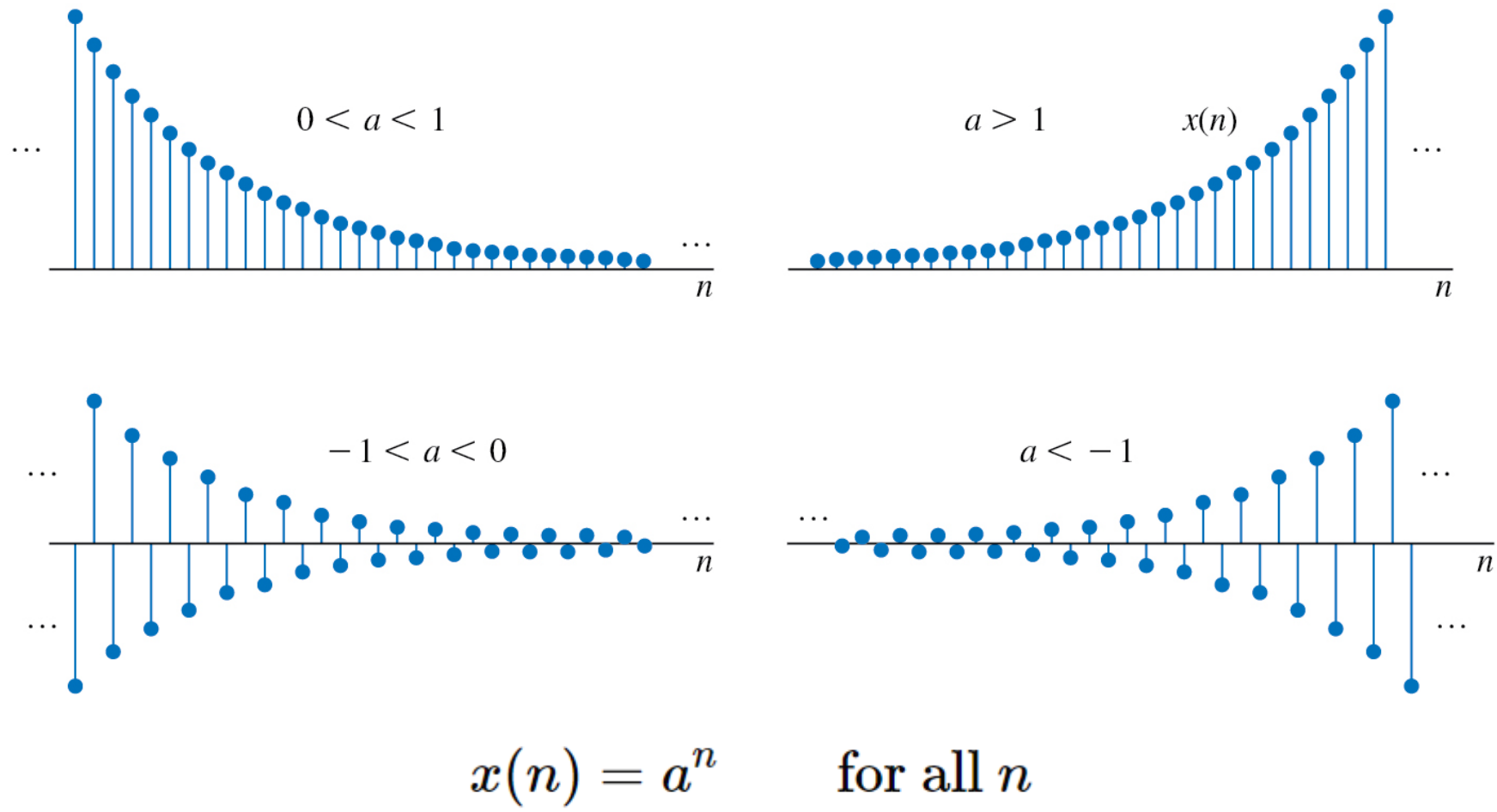


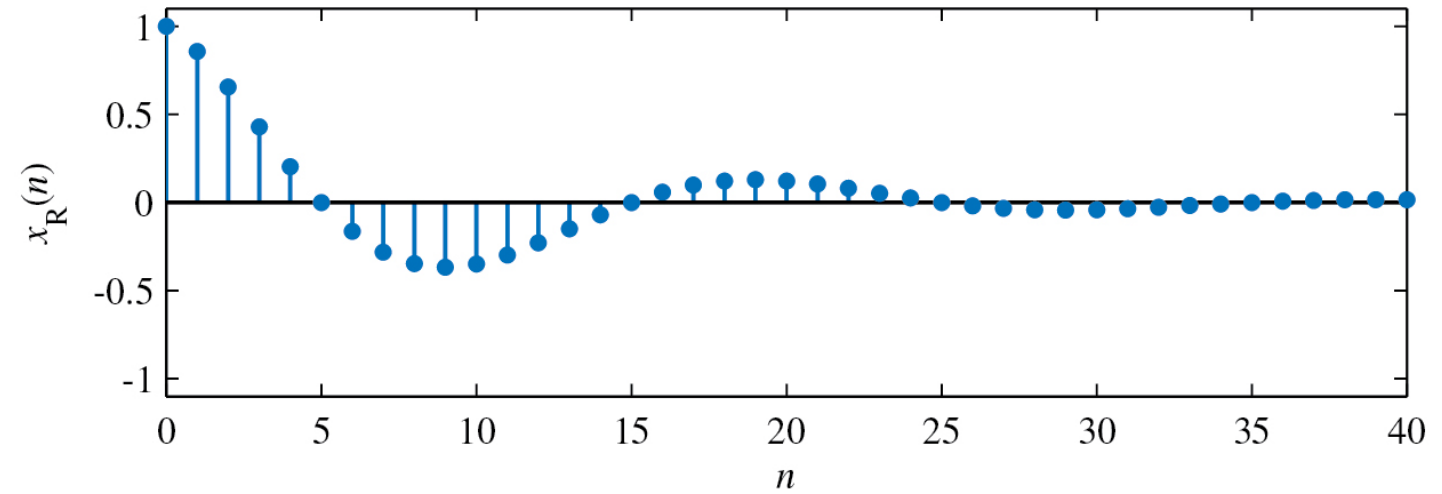
Figure 2.1.6 Graph of the real and imaginary components of a complex-valued exponential signal.

$$x(n) = a^n \quad \text{for all } n$$

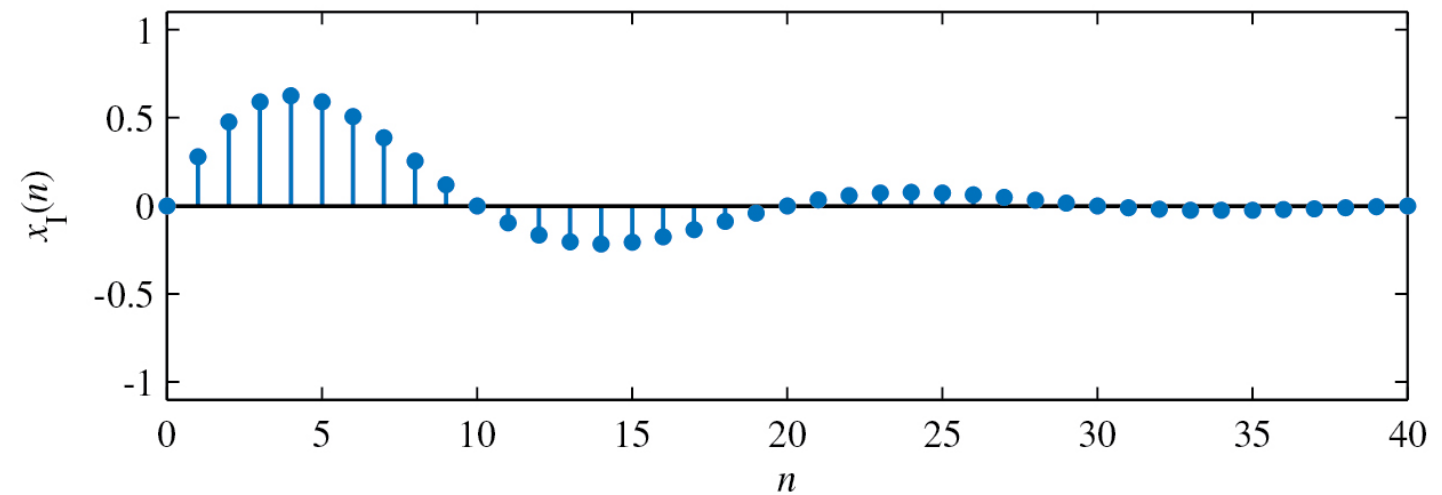
$$a \equiv r e^{j\theta}$$

$$\begin{aligned} x(n) &= r^n e^{j\theta n} \\ &= r^n (\cos \theta n + j \sin \theta n) \end{aligned}$$

Figure assumes $r = 0.09$, $\theta = \pi/10$

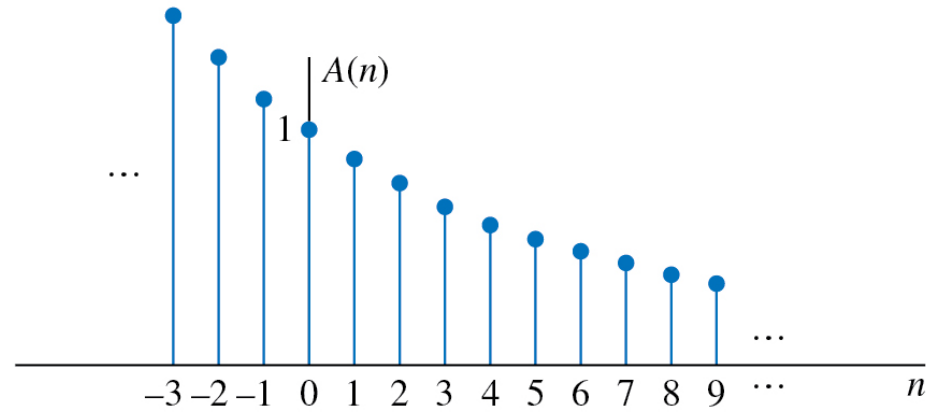


(a)

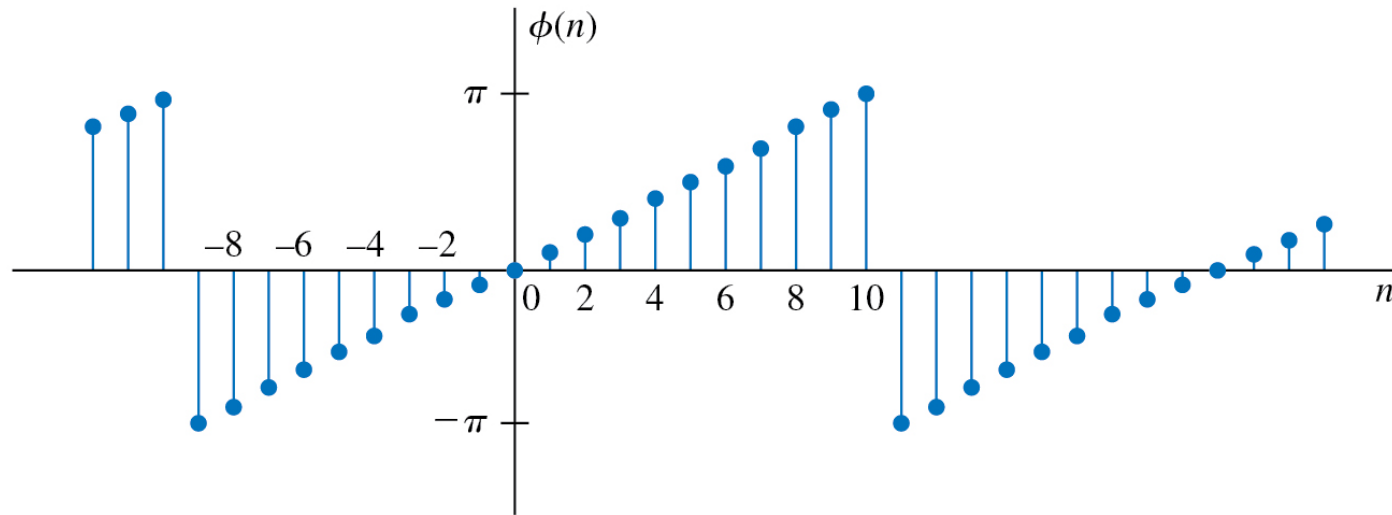


(b)

Figure 2.1.7 Graph of amplitude and phase function of a complex-valued exponential signal: (a) graph of $A(n) = r^n$, $r = 0.9$; (b) graph of $\phi(n) = (\pi/10)n$, modulo 2π plotted in the range $(-\pi, \pi]$.



(a) Graph of $A(n) = r^n$, $r = 0.9$



(b) Graph of $\phi(n) = \frac{\pi}{10}n$, modulo 2π plotted in the range $(-\pi, \pi)$

Simple Manipulation of Discrete-Time Signals

- Time shift (delay or advance): $x(n-k)$ where k is an integer.
 - If $k > 0$, delay; if $k < 0$, advance.
- Folding or reflection: $x(-n)$
 - Operations of folding and time shifting are NOT commutative.
- Time scaling or down-sampling: $x(\mu n)$

Figure 2.1.9 Graphical representation of a signal, and its delayed and advanced versions.

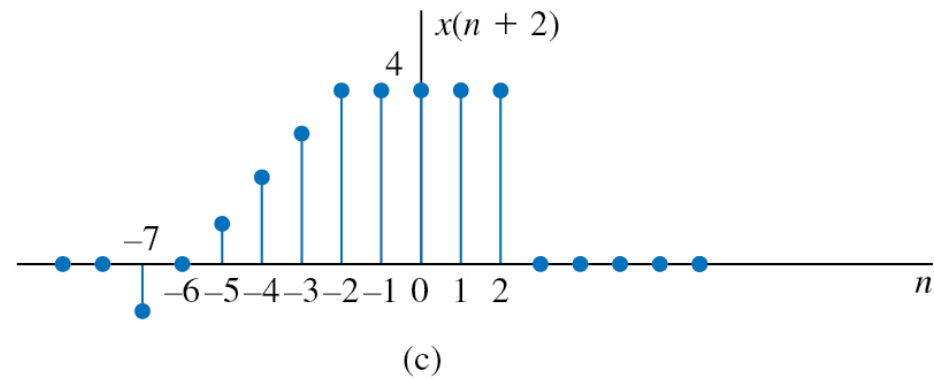
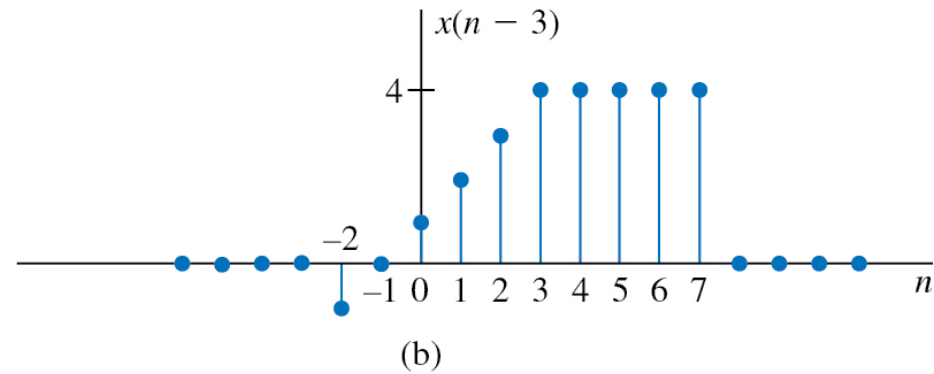
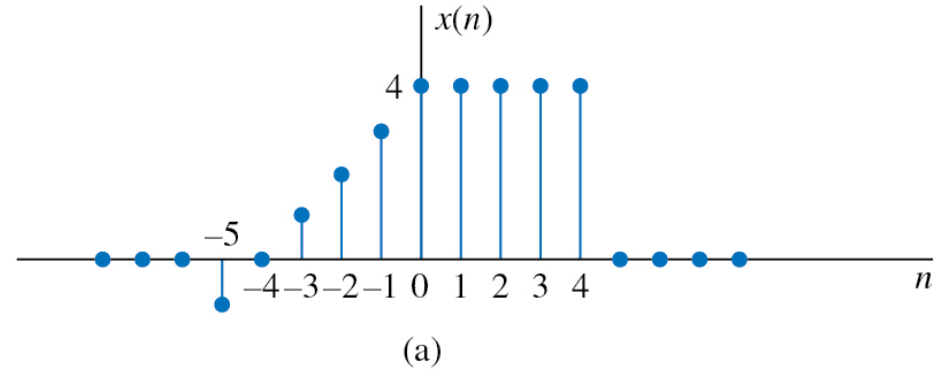
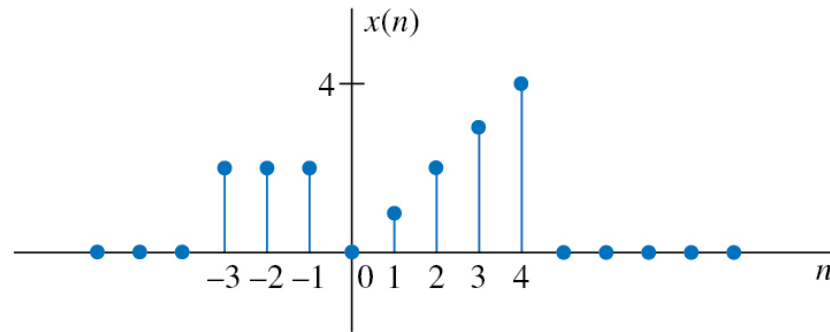
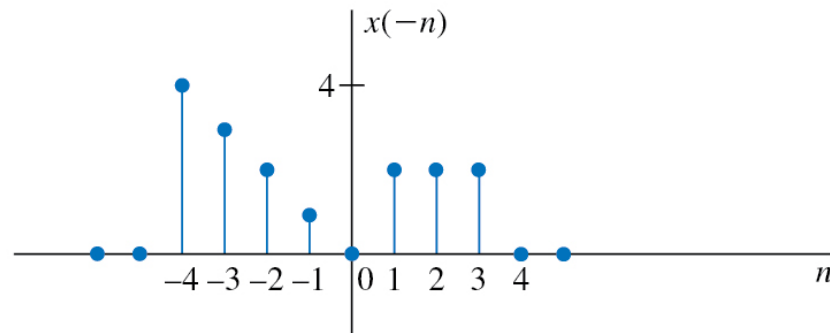


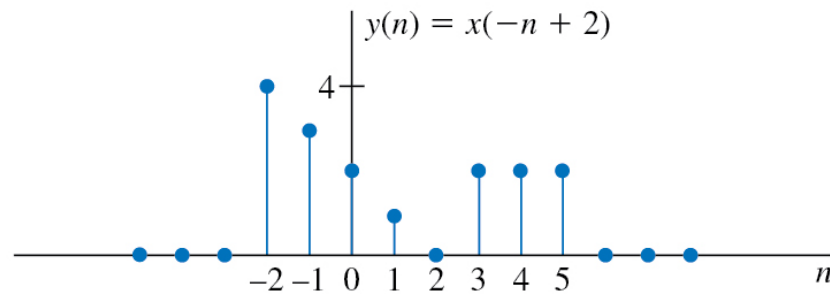
Figure 2.1.10 Graphical illustration of the folding and shifting operations.



(a)

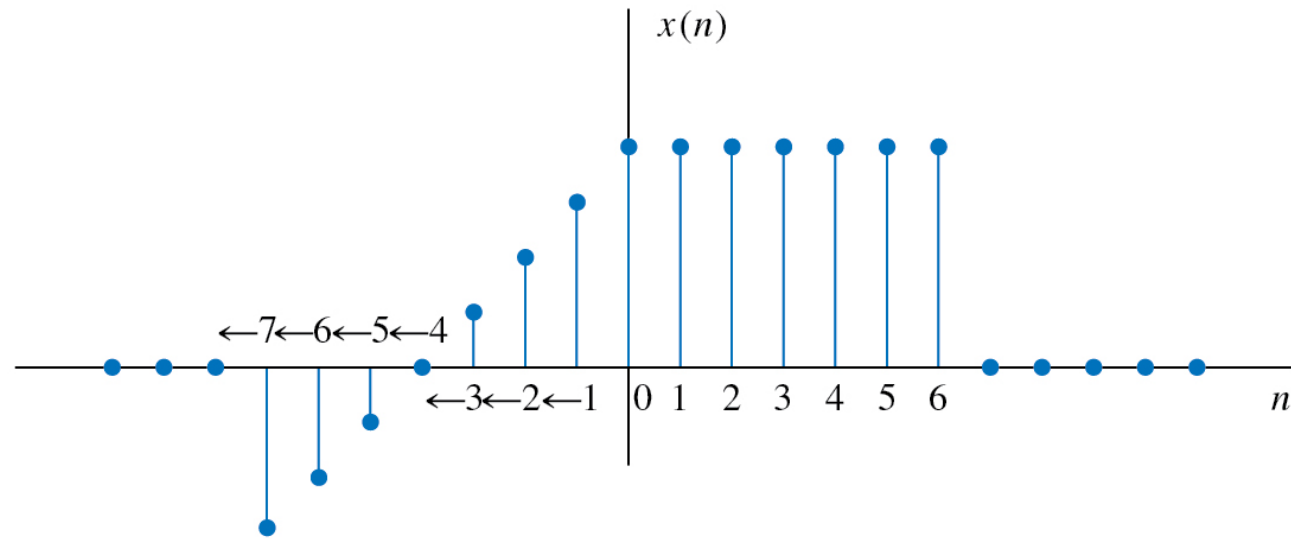


(b)

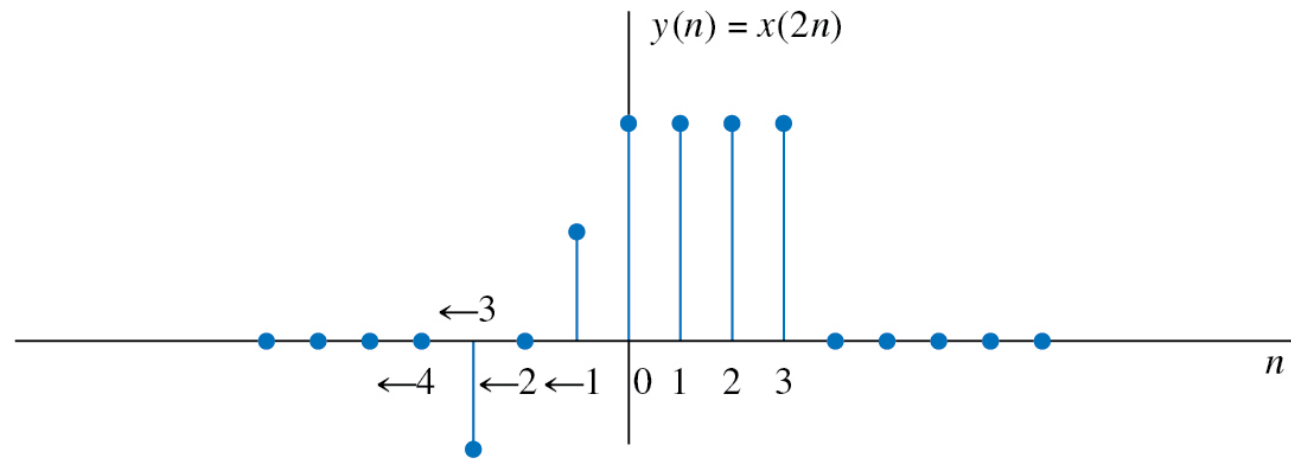


(c)

Figure 2.1.11 Graphical illustration of down-sampling operation.



(a)



(b)

Amplitude modifications

- Addition $y(n) = x_1(n) + x_2(n), \quad -\infty < n < \infty$
- Multiplication $y(n) = x_1(n)x_2(n), \quad -\infty < n < \infty$
- Amplitude scaling $y(n) = Ax(n), \quad -\infty < n < \infty$

Energy vs. Power Signals

- Energy E of a signal $x(n)$:

$$E \equiv \sum_{n=-\infty}^{\infty} |x(n)|^2$$

- **Energy signal:** defined as a signal for which E is finite: $0 < E < \infty$

- Average power of a signal $x(n)$:

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N + 1} \sum_{n=-N}^N |x(n)|^2$$

- If E is finite, $P = 0$. (for energy signals)
 - **Power signal:** defined as a signal for which P is finite and nonzero: $0 < P < \infty$

Periodic vs. Aperiodic Signals

- Periodic signal:

$$x(n + N) = x(n) \text{ for all } n$$

- The smallest value of N (>0) is called the (fundamental) period.
- Aperiodic (or nonperiodic) if no such N exists.

- Example- Sinusoidal signal: $x(n) = A \sin 2\pi f_0 n$

- If f_0 is a rational number (k & N are integers): $f_0 = \frac{k}{N}$

- Is a periodic signal an energy signal? A power signal? How can we simplify the calculation of the power?

Symmetric (even) vs. Antisymmetric (odd)

- Symmetric or even signal: $x(-n) = x(n)$
- Antisymmetric or odd signal: $x(-n) = -x(n)$
- Any signal can be written as the sum of even and odd components:

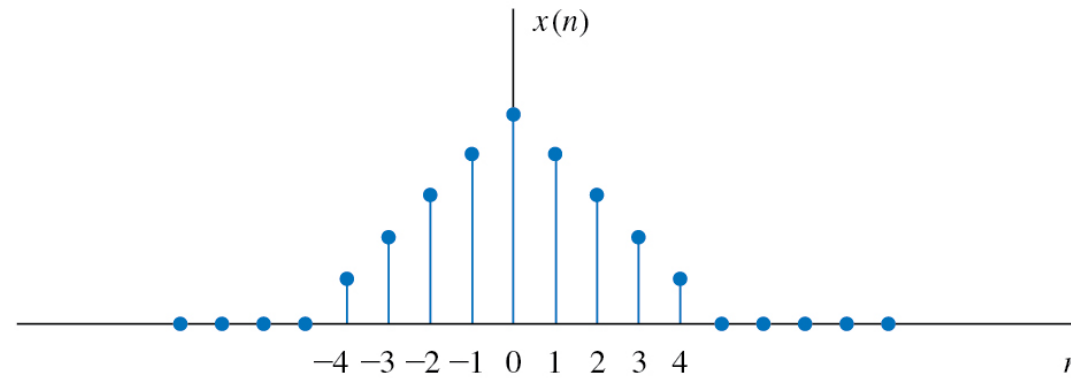
$$x(n) = x_e(n) + x_o(n)$$

where:

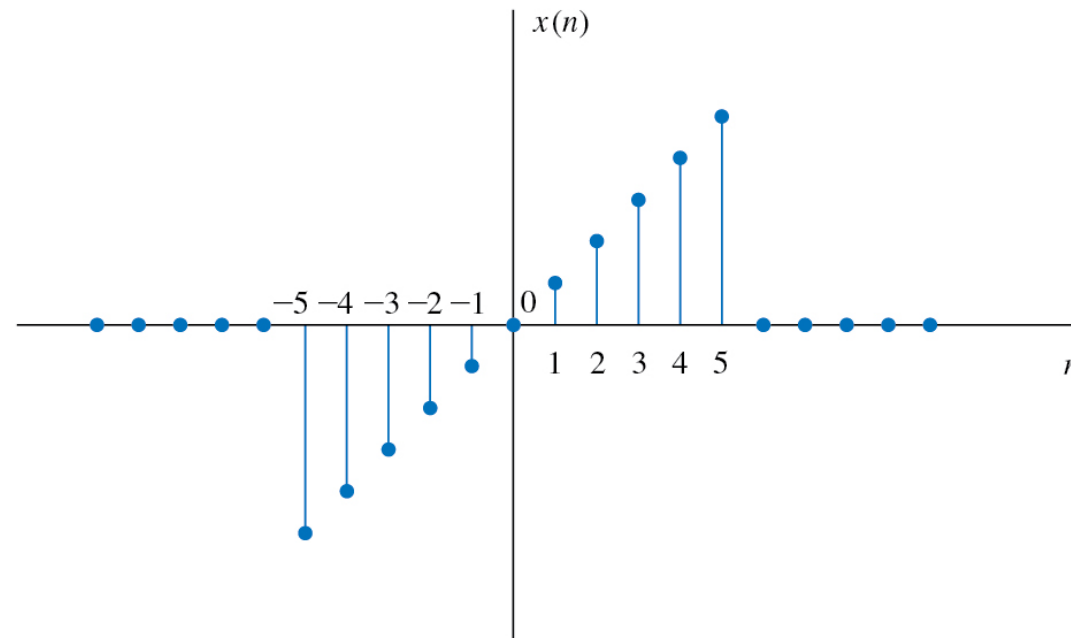
$$x_e(n) = \frac{1}{2}[x(n) + x(-n)]$$

$$x_o(n) = \frac{1}{2}[x(n) - x(-n)]$$

Figure 2.1.8 Example of even (a) and odd (b) signals.



(a)



(b)

Summary

References

- [DSP] Proakis, J.G. and Manolakis, D.G., Digital Signal Processing: Principles, Algorithms, and Applications, Pearson, 5th ed., 2022.

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